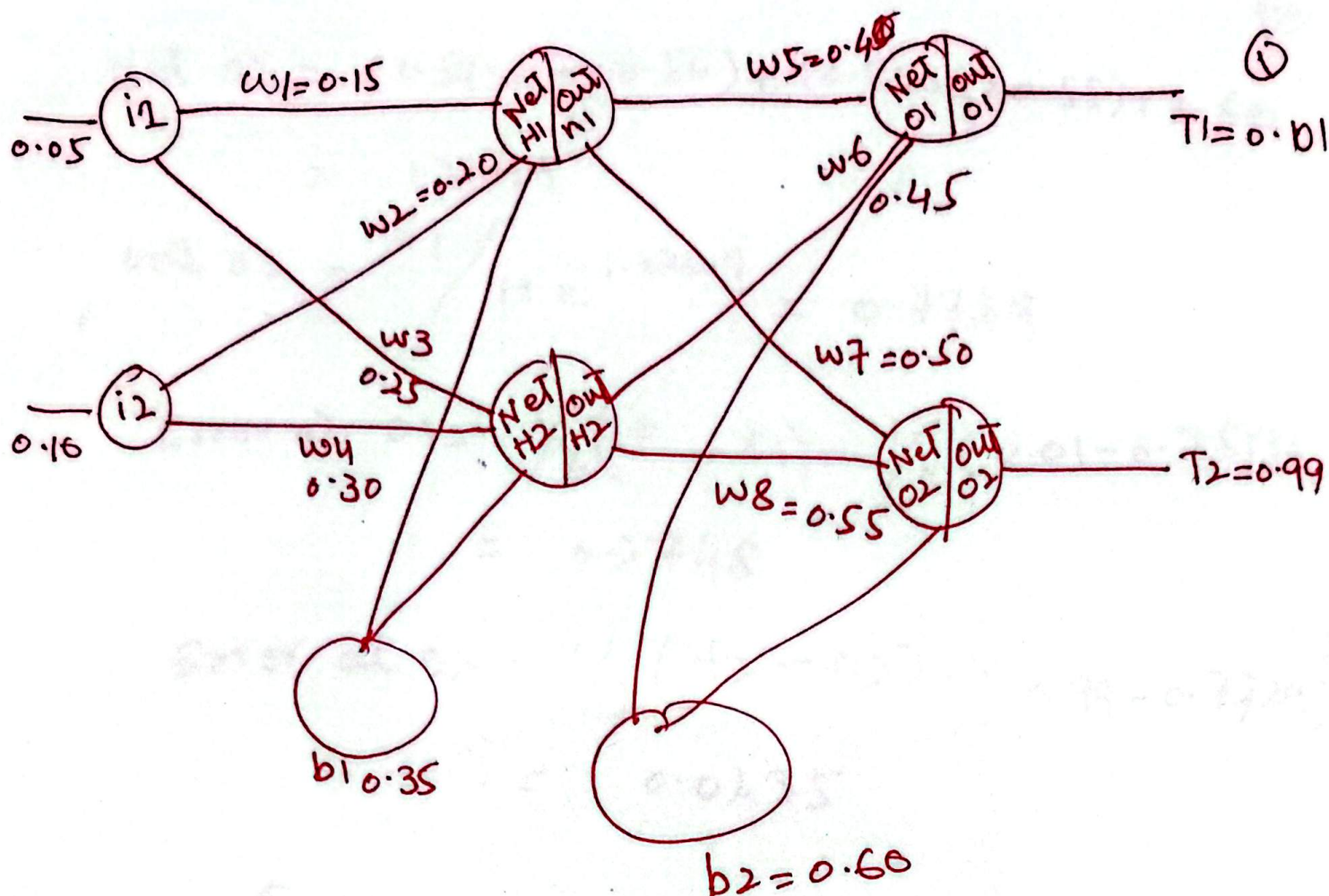




$$\text{Net } h_1 = (0.05 * 0.15) + (0.10 * 0.20) + 0.35 = 0.3775$$

$$\text{out } h_1 = \frac{1}{1 + e^{-\text{Net } h_1}} = \frac{1}{1 + e^{-0.3775}} = 0.59327$$

$$\text{Net } h_2 = (0.05 * 0.25) + (0.10 * 0.30) + 0.35 = 0.3925$$

$$\text{out } h_2 = \frac{1}{1 + e^{-0.3925}} = 0.5968$$

$$\text{Net } o_1 = (0.59327 * 0.40) + (0.5968 * 0.45) + 0.60 = 1.1059$$

$$\text{out } o_1 = \frac{1}{1 + e^{-1.1059}} = 0.75136$$

(2)

$$\text{Net } o_2 = (0.59327 \times 0.50) + (0.5968 \times 0.55) + 0.60$$

$$= 1.2249$$

$$\text{out } o_2 = \frac{1}{1 + e^{-1.2249}} = 0.7729$$

$$\text{Error at } o_1 = \frac{1}{2}(t_1 - o_1) = \frac{1}{2}(0.01 - 0.75136)^2$$

$$= 0.2748$$

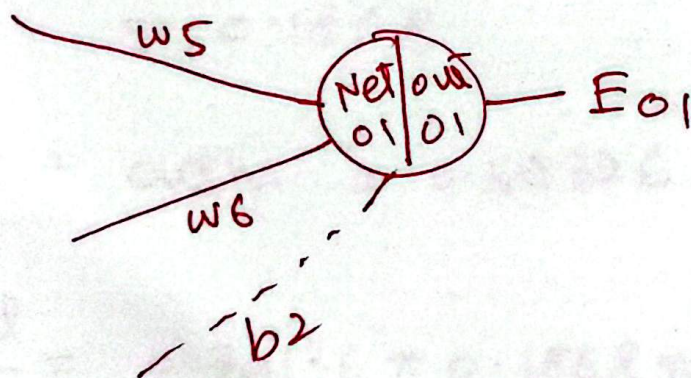
$$\text{Error at } o_2 = \frac{1}{2}(t_2 - o_2) = \frac{1}{2}(0.99 - 0.7729)^2$$

$$= 0.0235$$

$$E_{\text{Total}} = 0.2748 + 0.0235$$

$$= 0.2983$$

update w_5



$$\frac{\partial E_{\text{total}}}{\partial w_5} = \frac{\partial E_{\text{total}}}{\partial \text{out } o_1} * \frac{\partial \text{out } o_1}{\partial \text{net } o_1} * \frac{\partial \text{net } o_1}{\partial w_5}$$

$$\frac{\partial E_{total}}{\partial out_{01}} = out_{01} - Target_{01} \quad (3)$$

$$= 0.75136 - 0.01 = 0.74136$$

~~$$\frac{\partial out_{01}}{\partial net_{01}} = \frac{\partial}{\partial x} \left(\frac{1}{1+e^{-x}} \right)$$~~

$$= \frac{\partial}{\partial net_{01}} (\text{sigmoid}(net_{01}))$$

$$= \text{sigmoid}'(net_{01}) = (1 + e^{-net_{01}})^{-1}$$

$$\frac{\partial}{\partial net_{01}} = -1 \cdot (1 + e^{-net_{01}})^{-2} \cdot \frac{d}{dx} (1 + e^{-net_{01}})$$

$$\frac{\partial out_{01}}{\partial net_{01}} = out_{01} (1 - out_{01})$$

$$= 0.75136 (1 - 0.75136)$$

$$= 0.1868$$

$$\frac{\partial net_{01}}{\partial w_5} = out_{in} = 0.59326$$

$$\frac{\partial E_{total}}{\partial w_5} = 0.74136 * 0.1868 * 0.59326$$

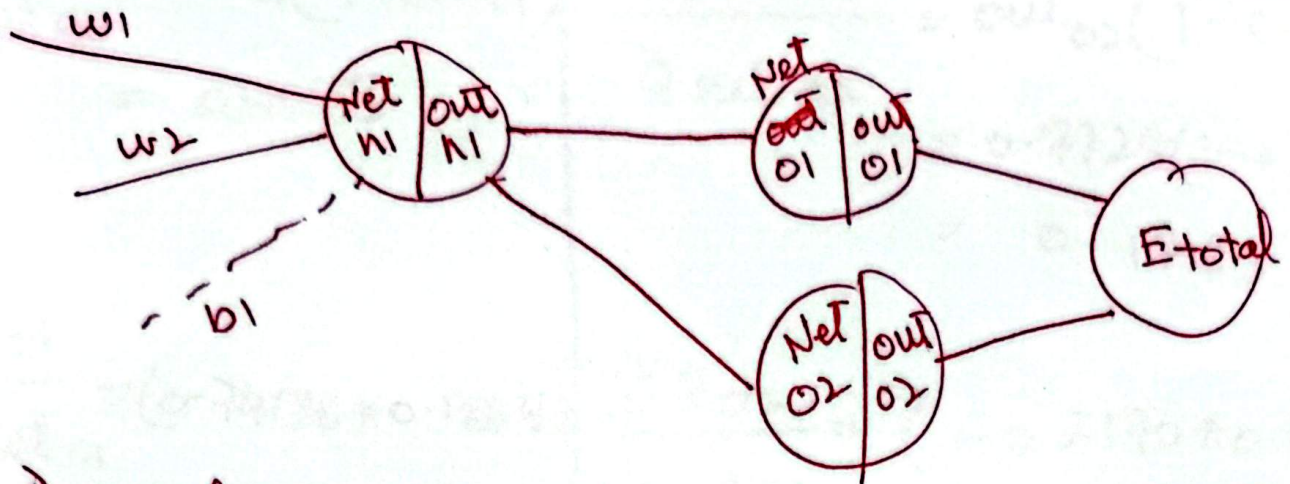
$$= 0.0822$$

$$w_5^* = w_5 - \alpha \frac{\partial E_{total}}{\partial w_5} = 0.4 - 0.6 * 0.0822$$

$$= 0.3504$$

update w_1

(4)



$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial out_{h1}} * \frac{\partial out_{h1}}{\partial net_{h1}} * \frac{\partial net_{h1}}{\partial w_1}$$

$$\frac{\partial E_{total}}{\partial out_{h1}} = \frac{\partial E_{o1}}{\partial out_{h1}} + \frac{\partial E_{o2}}{\partial out_{h1}}$$

Arrows indicate the flow of derivatives from the output node to the two hidden nodes.

$$\frac{\partial E_{o1}}{\partial net_{o1}} * \frac{\partial net_{o1}}{\partial out_{h1} (w_5)}$$

$$\frac{\partial E_{o1}}{\partial out_{o1}} * \frac{\partial out_{o1}}{\partial net_{o1}}$$

$$(out_{o1} - target_{o1})$$

$$= 0.75136 - 0.01$$

$$= 0.74136$$

$$\frac{\partial E_{o2}}{\partial net_{o2}} * \frac{\partial net_{o2}}{\partial out_{h1} (w_7)}$$

$$\frac{\partial E_{o2}}{\partial out_{o2}} * \frac{\partial out_{o2}}{\partial net_{o2}}$$

$$(out_{o2} - target_{o2})$$

$$(0.7729 - 0.99)$$

$$= -0.2170$$

(5)

$$\frac{\delta out_{01}}{\delta net_{01}} = out_{01}(1-out_{01})$$

$$= 0.1868$$

$$\frac{\delta out_{02}}{\delta net_{02}} = out_{02}(1-out_{02})$$

$$= 0.7729(1-0.7729)$$

$$= 0.1755$$

$$\frac{\delta E_{01}}{\delta net_{01}} = (-0.74136 * 0.1868)$$

$$= (-0.1385)$$

$$\frac{\delta E_{02}}{\delta net_{02}} = (-0.2170 * 0.1755)$$

$$= (-0.038)$$

$$\frac{\delta net_{01}}{\delta out_{h1}} = 0.4$$

$$\frac{\delta net_{02}}{\delta out_{h1}} = 0.50$$

$$\frac{\delta E_{01}}{\delta out_{h1}} = 0.1385 * 0.4$$

$$= 0.055$$

$$\frac{\delta E_{02}}{\delta out_{h1}} = (-0.038 * 0.50)$$

$$= -0.019$$

$$\frac{\delta E_{total}}{\delta out_{h1}} = 0.055 + (-0.019)$$

$$= 0.03635$$

$$\frac{\delta out_{h1}}{\delta net_{h1}} = out_{h1}(1-out_{h1})$$

$$= 0.7536(1-0.7536)$$

$$= 0.2413$$

$$\frac{\delta net_{h1}}{\delta w_1} = input_{01} = 0.05$$

$$\frac{\partial E_{total}}{\partial w_1} = \frac{\partial E_{total}}{\partial out_{h_1}} * \frac{\partial out_{h_1}}{\partial net_{h_1}} * \frac{\partial net_{h_1}}{\partial w_1} \quad (6)$$

$$= 0.03635 * 0.2413 * 0.05$$

$$= 0.0004385$$

$$w_1^* = w_1 - \alpha * \frac{\partial E_{total}}{\partial w_1}$$

$$= 0.15 - 0.60 * 0.0004385$$

$$= 0.1497$$